



A tool to predict melting glaciers

NASA developed a new tool that can predict which glaciers and ice sheets can flood your city as the Earth warms up. <http://digit.in/NASATool>



First observable interstellar asteroid

ESO's Very Large Telescope (VLT) was used to measure Oumuamua's orbit, brightness and colour more accurately. <http://digit.in/IAsT>

Of course, quantum physicists are used to isolating quantum effects and studying them at near absolute zero temperatures (-273 degrees C), while biology is done at relatively hot temperatures with a billion different things happening at once... so it's certainly not an easy field to study.

IS IT NEW?

Heck no! As early as 1944, Erwin Schrödinger, the famous Austrian physicist, known more for his cat thought experiment than anything else, published a book called *What is Life?* In it he looked at life (or biology in general) through the lens of a physicist. Much before DNA was discovered, in this book he suggested that perhaps there was some storage mechanism of information inside all living things which told them what they were and how to grow, etc. He thought it might be a special crystal of some type which contained information inside its chemical covalent bonds. Needless to say, when James Watson and Francis Crick discovered DNA's structure in 1953, using an amazing photograph taken by Rosalind Franklin (see our special book for more details), they listed Schrödinger's book as an inspiration.

THE LIES YOU'RE TOLD

You might think it's all been discovered, and there's nothing really left to work on in biology. Nothing could be further from the truth. Take DNA for example: we all know it's a chemical compound, we even know the shape, the base pairs, the structure, and although expensive, the Human Genome Project (HGP) spent billions of dollars sequencing a whole human genome. Except, *they didn't*.

Most of us think the DNA analysis that crime labs do are merely a case of popping some DNA into a machine, which reads the whole sequence from start to finish, and then compare that to a suspect's DNA and then we go "Aha! Caught you!", if it's a match.

Nothing is further from the truth. We all know DNA is made up of the "letters" A C G T. Here, A = Adenine,

which is actually $C_5H_5N_5$, Cytosine is $C_4H_5N_3O$, Guanine is $C_5H_5N_5O$, and Thymine is $C_5H_6N_2O_2$. Now, when we heard that the HGP had sequenced the human genome, we tend to think of it as the entire genome of a human was read from start to finish, it wasn't. Human DNA is 3 billion base pairs long – in sperm and eggs with 23 chromosomes; and 6 billion pairs long in non-sex cells, or 46 chromosomes. This makes it impossible for any technology we have currently to just read the entire DNA strand. Instead, the DNA is broken into chunks, and then the bits and pieces read, and then arranged together to form the read out. Think of it like a jigsaw puzzle, except this one has a hundred million pieces! When the HGP was completed in 2003, it still had gaps in the DNA that weren't sequenced. This is because although all the chunks of DNA were read (think of it as having all the pieces of the jigsaw), there were a few patches where everything seemed to fit, and thus, we couldn't tell what was the actual piece that went there. Think of this as a clear blue sky in a jigsaw where many pieces are almost identically shaped, and you really don't know which one goes where in order to be able to complete that featureless patch of blue sky. (An oversimplification, obviously).

In essence, there is a lot about DNA that we still don't know but can guess, and even more that we have no clue about.

The reason we took the DNA detour is to establish that we really don't know a lot about things laymen like us think scientists know everything about... and it's not just limited to DNA.

BIRDS

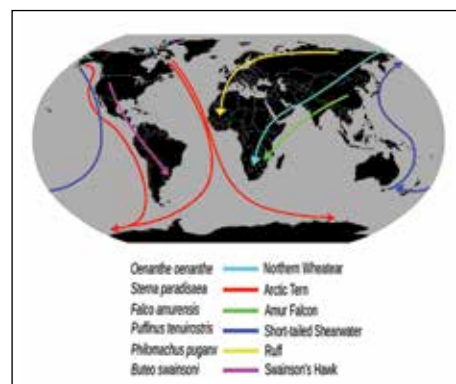
For decades we've known that birds migrate to and from very specific places on earth, down to a meter accuracy on journeys that can last thousands of kilometres, and in varying weather conditions, and even when flying at night. So how do they do it? We've set

up radar stations, GPS satellites and use advanced techniques to navigate in our high tech aircraft, while birds have been doing it successfully for far longer than we've even been human.

In school, we're all told that birds can sense the magnetic field of the earth, and that's how they migrate. Something inside them allows them to feel the extremely weak lines of magnetism, and follow them due north or south, depending on the season. This part was never explained to us though – not if you had a good science teacher who didn't speculate, because we just don't know how they do it. No theory was able to properly explain how birds are able to sense magnetic fields.

SO HOW DO THEY DO IT?

That's where quantum biology comes in. A whole lot of researchers have done experiments and found that birds need light to hit their eyes to be able to sense the magnetic field. Not even sunlight, but even the dim lighting of night time works. There were a ton of experi-



Examples of the vast distances some birds migrate

ments run to settle on this, including putting hoods on birds to cover their eyes, and playing with magnetic fields, and it became obvious that light had something to do with a bird being able to sense magnetic fields. Now, even more research has shown that there are cells in all birds' eyes which contain a light sensitive protein called cryptochrome, and these proteins are activated by blue or green light, and create what is known as a radical pair.